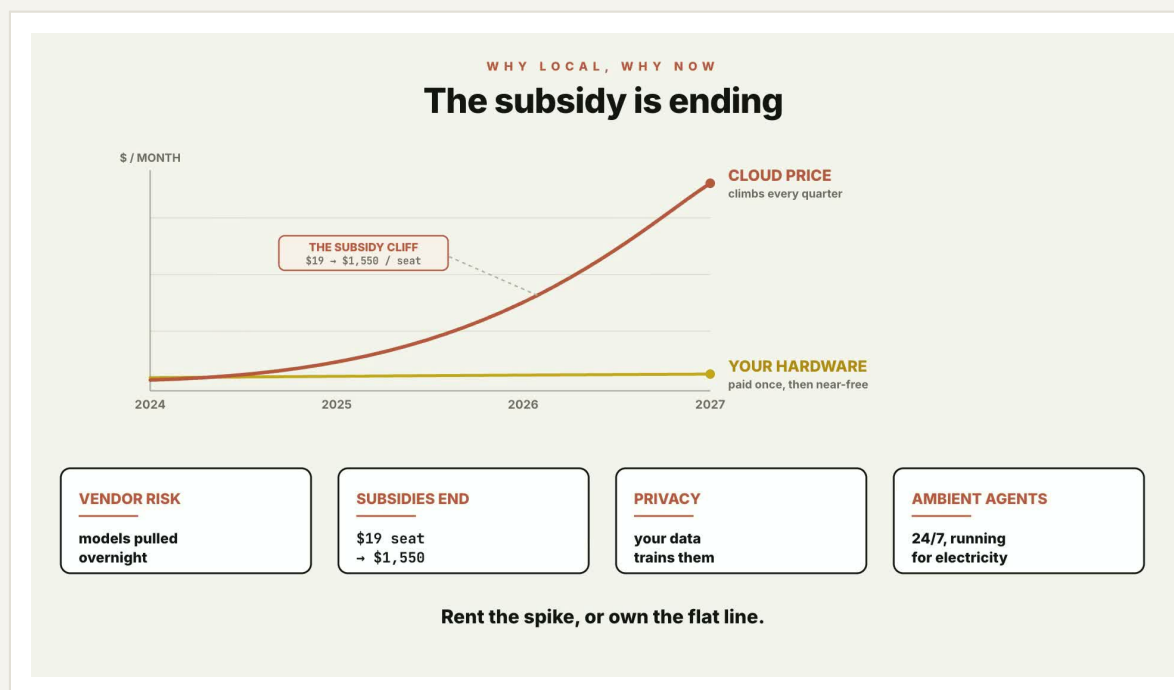


OPEN SOURCE AI CRASH COURSE

The complete companion guide. Concepts, the five-floor stack, models, quantization, harnesses, and the path of least resistance to your own local command center.

EARLY AI-DOPTERS · skool.com/earlyaidopters

Rent Goes Up. Owned Hardware Stays Flat.



VENDOR RISK IS REAL

Frontier models have vanished from public hands overnight before, and some never arrive at all. Whatever your daily workflow leans on can change in the snap of a finger.

THE SUBSIDY IS ENDING

VC money currently turns \$100 to \$200 plans into thousands of dollars of real usage. When these companies go public, the bill catches up with the ecosystem.

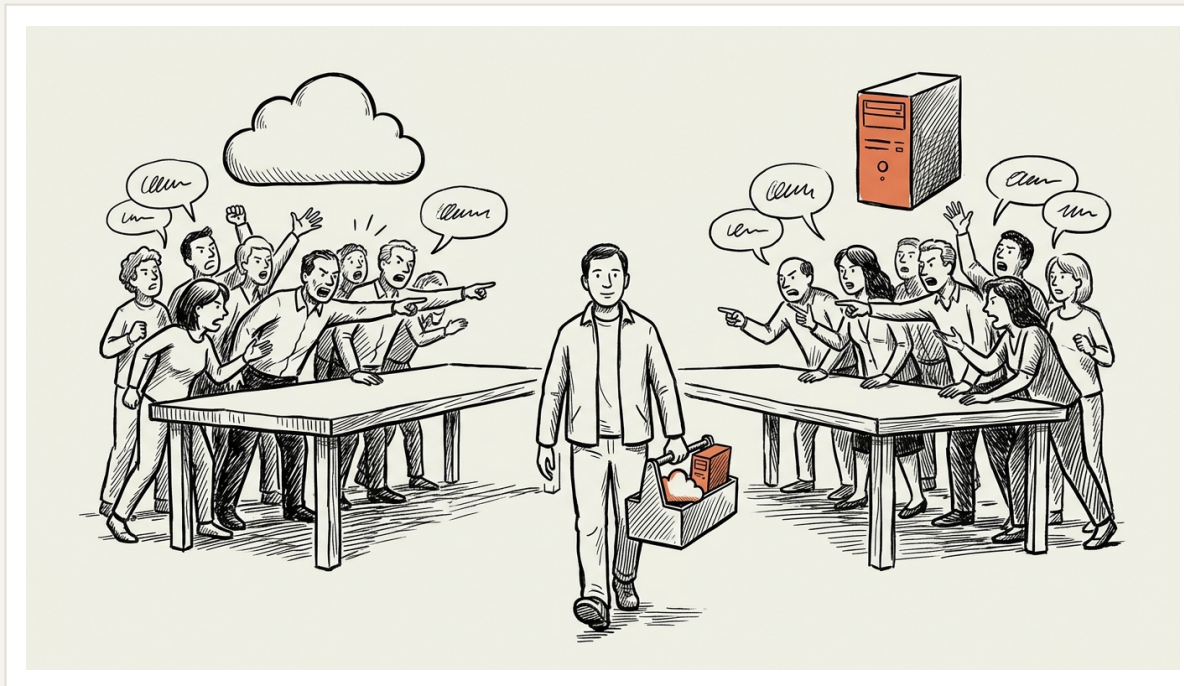
PRIVACY BY ARCHITECTURE

Personal data processed locally never touches a training pipeline. Analyze the sensitive material at home, then let cloud models work on the synthesis.

AMBIENT AGENTS

A 24/7 team of background agents doing the dirty work (file RAG, health checks, security sweeps) costs electricity, not API credits.

Two Camps, One Toolbox



BOTH CAMPS ARE WRONG

Team closed source calls local models toys. Team open source calls cloud users doomed. The working answer is in the middle: use both on purpose.

OPEN IS GOOD ENOUGH

Not the leading edge, and it doesn't need to be. For a large share of day-to-day work, current open models clear the bar and keep improving.

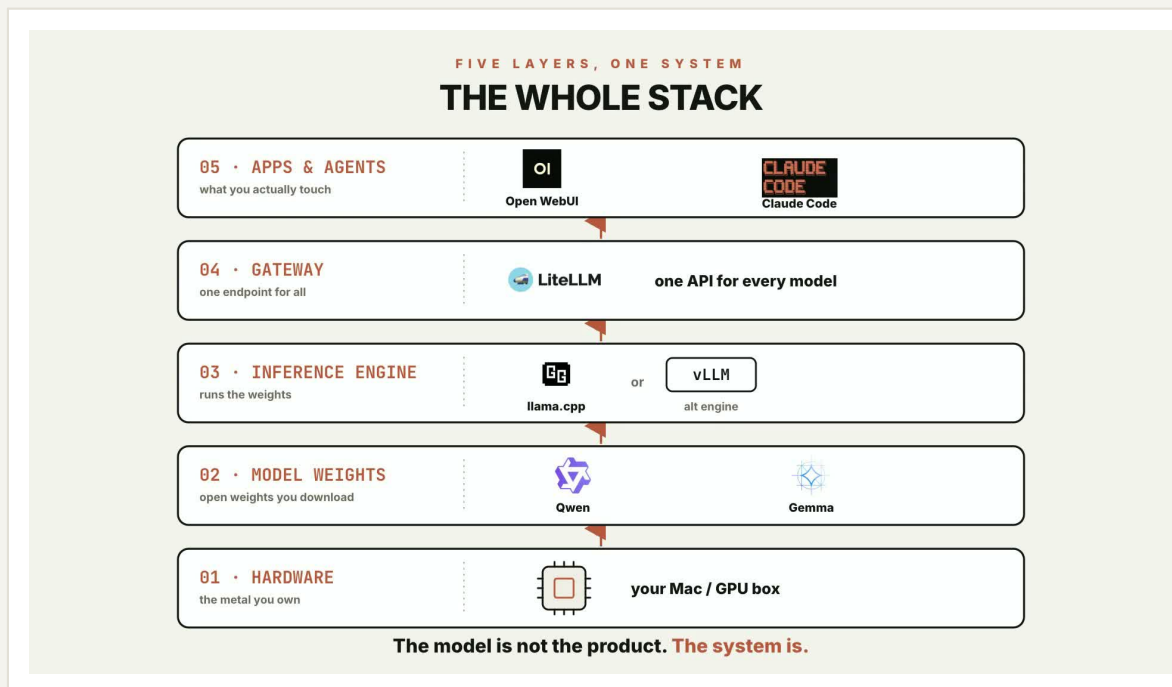
THE INSURANCE POLICY

Knowing how to use, enable, and host open models is a mental insurance policy against price hikes, access loss, and lock-in.

THE HONEST TRICK

Closed source builds the open source rig. Claude or Codex sets up, maintains, and bug-fixes your local stack so you never drown in the weeds.

Five Floors, Bottom to Top



1 • HARDWARE

Your machine decides what models run, at what capacity, at what speed. Everything above builds on this floor.

2 • MODELS

Open weight means the full weight file is public: you can see exactly how the model derives its answers, unlike closed APIs.

3 • INFERENCE ENGINE

llama.cpp is the bridge that turns your prompt into output. vLLM is the same idea built for team scale.

4+5 • GATEWAY & INTERFACE

LiteLLM is the plug adapter connecting any app to any model. On top: chat UIs like Open WebUI and harnesses that give the model limbs.

Same Cooking, Different Kitchen



COOKING FOR ONE

On a personal machine you are a party of one. llama.cpp is all you need to feed every interface you use.

THE INDUSTRIAL KITCHEN

vLLM exists for when many people hit the same models at once. Same recipes, industrial throughput.

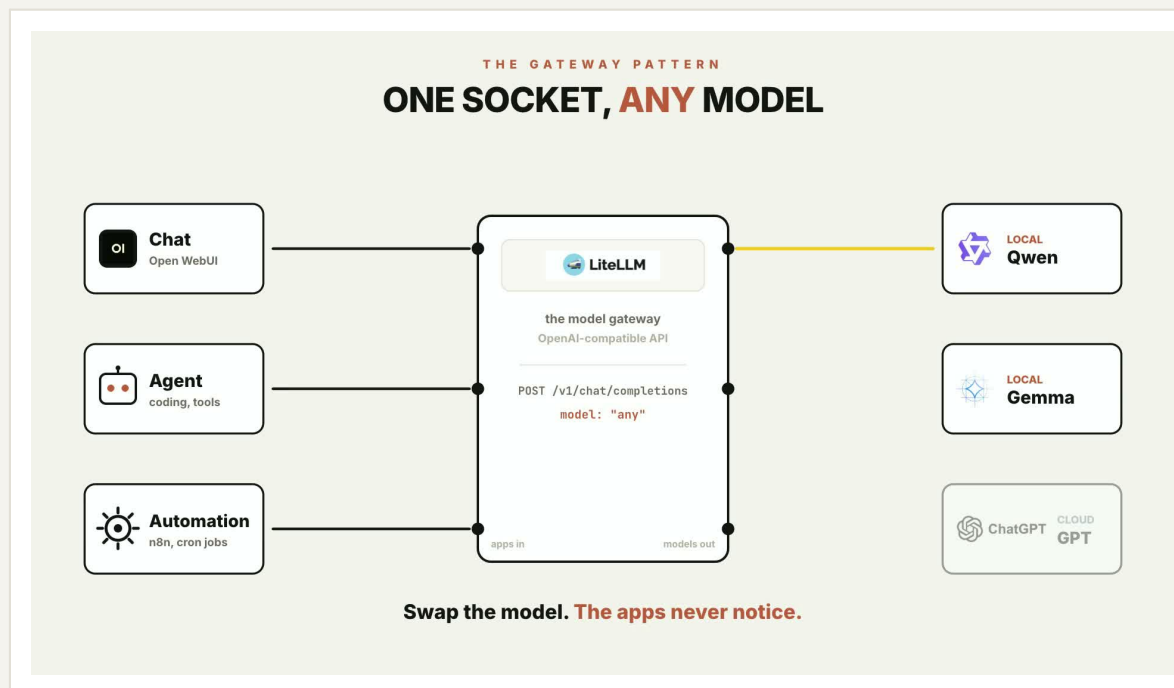
THE BEATER VS THE AUTOBAHN

A beater gets you from A to B alone. Force it onto a 300 mph highway and it breaks. Pick the engine for the load, not the brochure.

DON'T OVERTHINK IT

If you're serving yourself, the default engine underneath Ollama or LM Studio is already the right one.

Apps Plug Into the Board, Not the Model



ONE BACKEND, HUNDREDS OF MODELS

LiteLLM speaks the OpenAI-compatible API, so every tool that talks that dialect can talk to any model behind the board.

SWAPPING IS ONE CABLE

Moving from Qwen to Gemma is unplugging one cable and seating another. No rewiring your chat, agents, or automations.

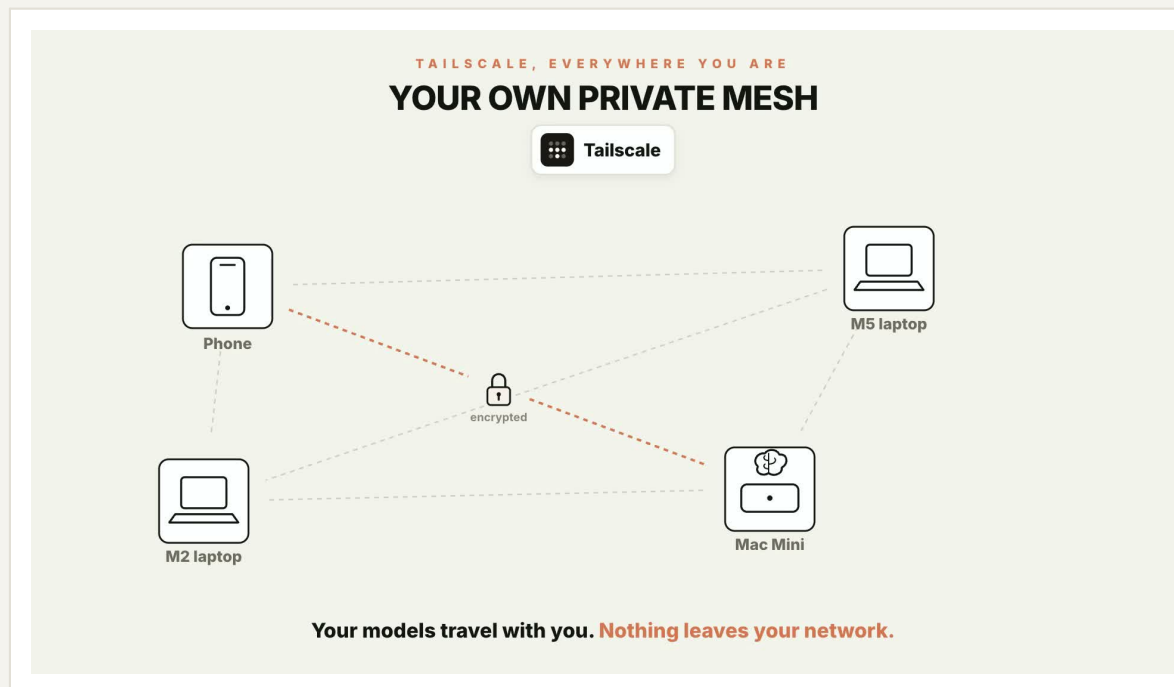
EVERYTHING ROUTES THROUGH IT

Chat UI, parallel coding agents, ambient agents, local n8n automations: all of them dial the same switchboard.

THE POINT IS NIMBLENESS

Models will keep leapfrogging each other monthly. The switchboard is what lets you ride that without rebuilds.

The Prompt Commutes, the Model Stays Home



A PRIVATE ENCRYPTED NETWORK

Tailscale joins your phone, laptops, and desktop into one mesh, as if they all sat in the same room. No port forwarding, no exposed server.

PHONE TO MAC AND BACK

Send a prompt from your phone: it travels encrypted to the machine running inference, and the answer commutes back. The model never leaves home.

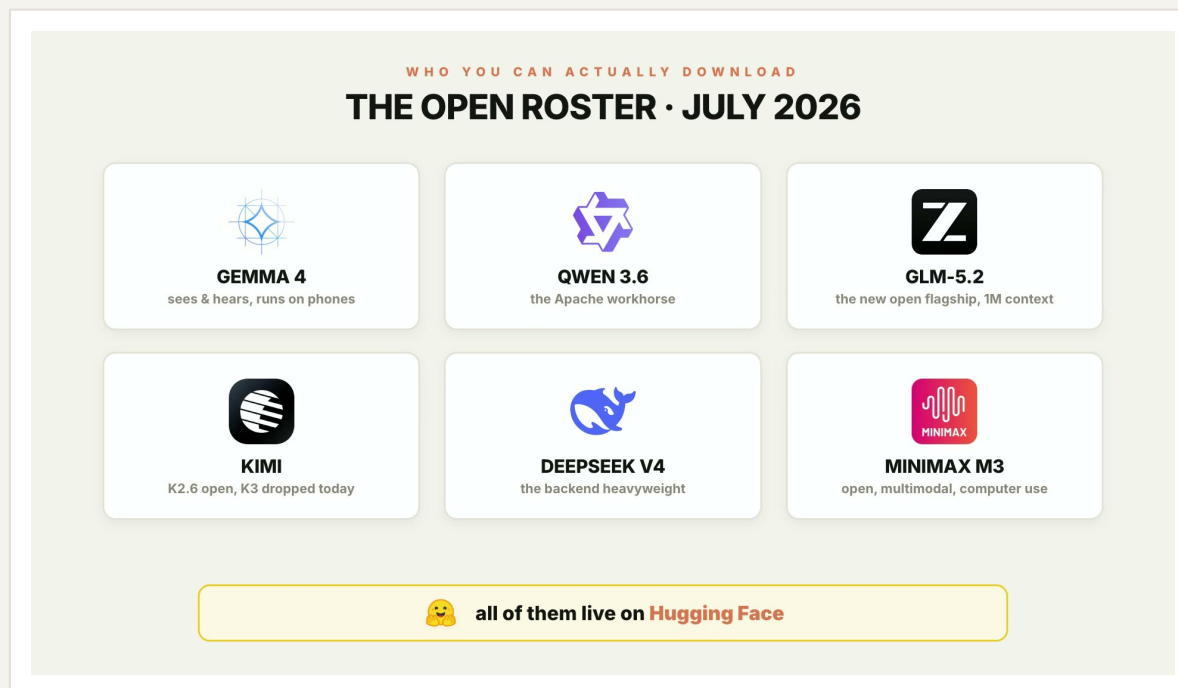
FREE FOR PERSONAL USE

The personal plan covers unlimited devices and up to six users. One Mini or five, same price: zero.

SHAREABLE ON PURPOSE

Invite a family member or teammate onto the mesh and your command center becomes a private, portable service.

Six Families You Can Download This Afternoon



THE QUIET PART OUT LOUD

The Chinese families (Qwen, GLM, Kimi) outperform much of the market, ship constantly, and now reach million-token context windows.

GEMMA & DEEPSEEK

Gemma is a capable all-rounder that can be oddly prohibitive. DeepSeek gave open source its breakout moment a few years ago.

MINIMAX FOR VOICE

Interesting alternative to commercial voice tools, with unique voice-cloning characteristics.

WORKHORSE PATTERN

Let closed source do detailed planning and review; offload mechanical tasks (renames, greps, small edits) to a local workhorse to save tokens and cost.

The App Store of Models



WHAT LIVES THERE

Vendor releases, quantized compressions, fine-tunes, distills: every permutation of open model, downloadable.

OPENROUTER SERVES, HF DELIVERS

OpenRouter lets you call a model over API. Hugging Face is where the actual weights come home to your disk.

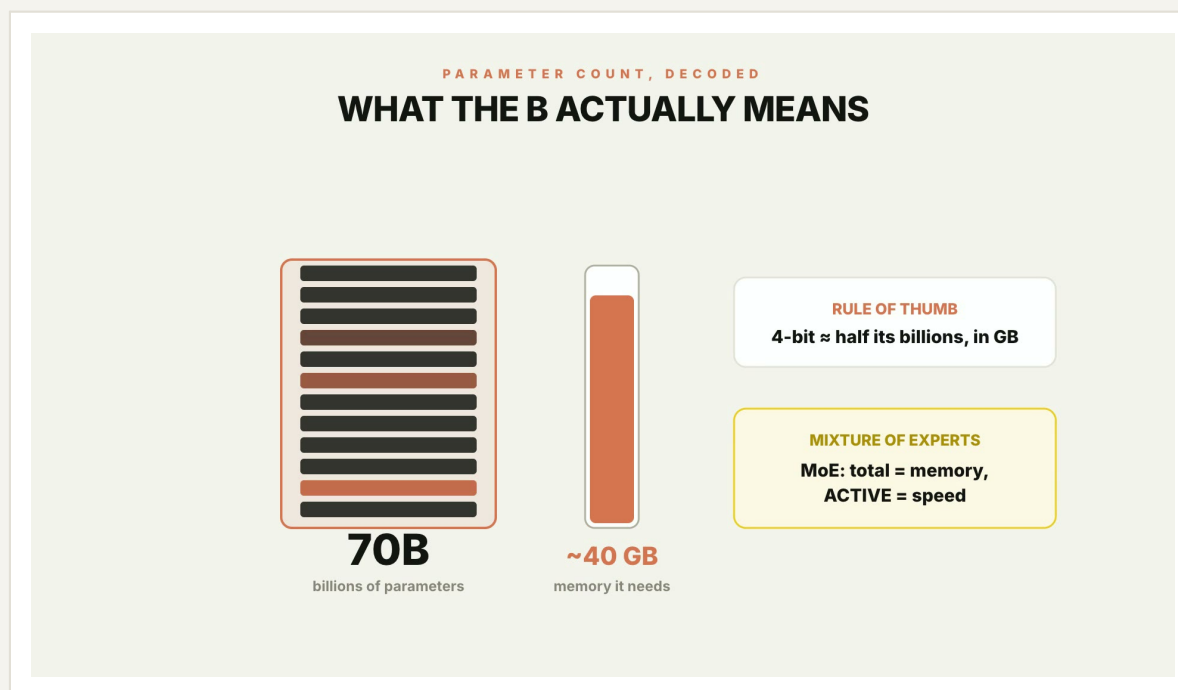
AUDITION BEFORE YOU BUY

Put \$10 in OpenRouter, test five models on ten of your real tasks, then only download the winners.

CHECK THE VARIANT

Most models ship in multiple formats. Grab the one tailored to your hardware (MLX on Apple Silicon, GGUF anywhere).

The B Is a Memory Bill



PARAMS ≈ CAPABILITY

Treat the billions count as a proxy for how capable a model is. Full Kimi K3 is ~2.8T parameters and would cost thousands a month to host at home.

THE DIVIDE-BY-TWO RULE

Parameters divided by two ≈ the GB of memory you need at 4-bit. A 70B model wants ~35-40GB; a 14B fits in 5-7GB.

USABLE MEMORY < BOX MEMORY

Your OS and apps eat RAM too. Don't budget the number printed on the box; budget what's actually free.

GOOD ENOUGH IS THE BAR

A 14B, 27B, or 50B is now sufficient for a surprising share of daily work, and that floor rises every quarter.

Open source isn't about picking a side. It's the mental insurance policy that keeps you nimble, informed, and a lot less afraid.

JPEG for Brains



RESOLUTION FOR MINDS

16-bit, 8-bit, 4-bit: like video resolution, each step down trades fidelity for size. 4K to 1080p is fine; below 1-bit is lobotomy territory.

THE RULE

Run the smallest quantization of the smallest parameter count that still passes YOUR tests. Nothing bigger earns its memory.

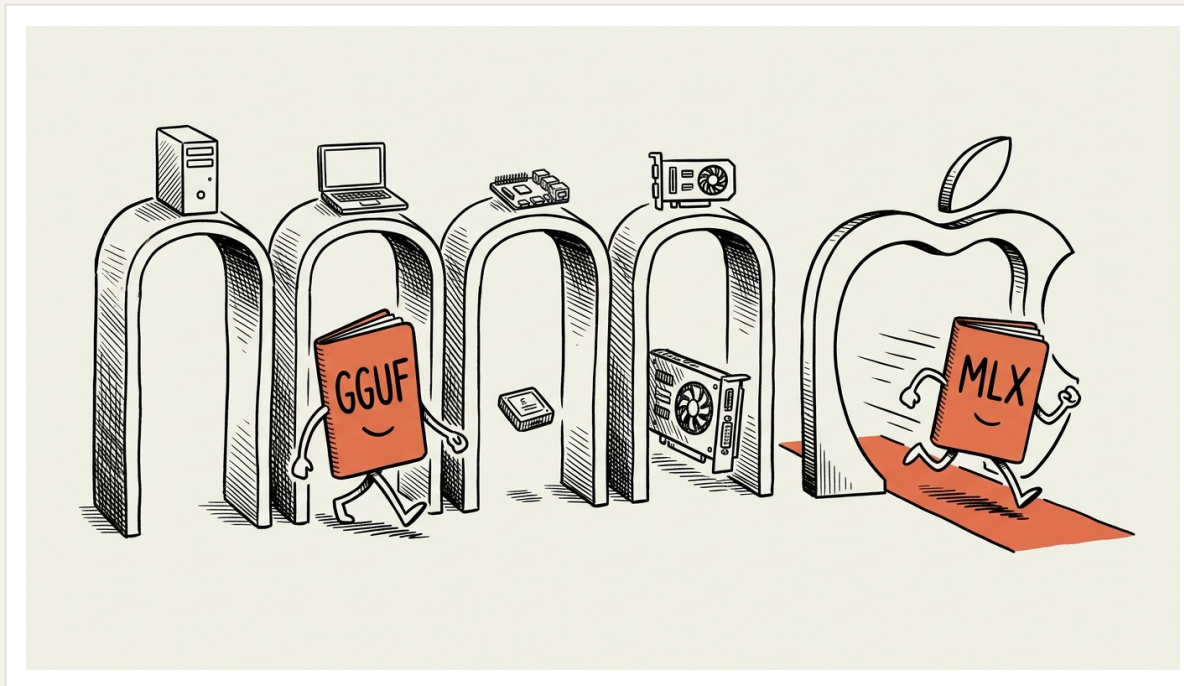
ALPHABET SOUP

The compression algorithm suffixes are implementation detail. You choose bits and parameter count; the suffix comes along.

WHY IT MATTERS

Smaller blocks flow faster through your memory pipe, which is exactly what tokens per second measures.

Two Passports, One Border



GGUF: THE UNIVERSAL PASSPORT

Works with practically everything. Nine times out of ten it runs out of the box with llama.cpp, Ollama, and LM Studio.

MLX: THE APPLE FAST-TRACK

Same model, optimized for Apple Silicon. Noticeably faster on a Mac because it's built for that architecture.

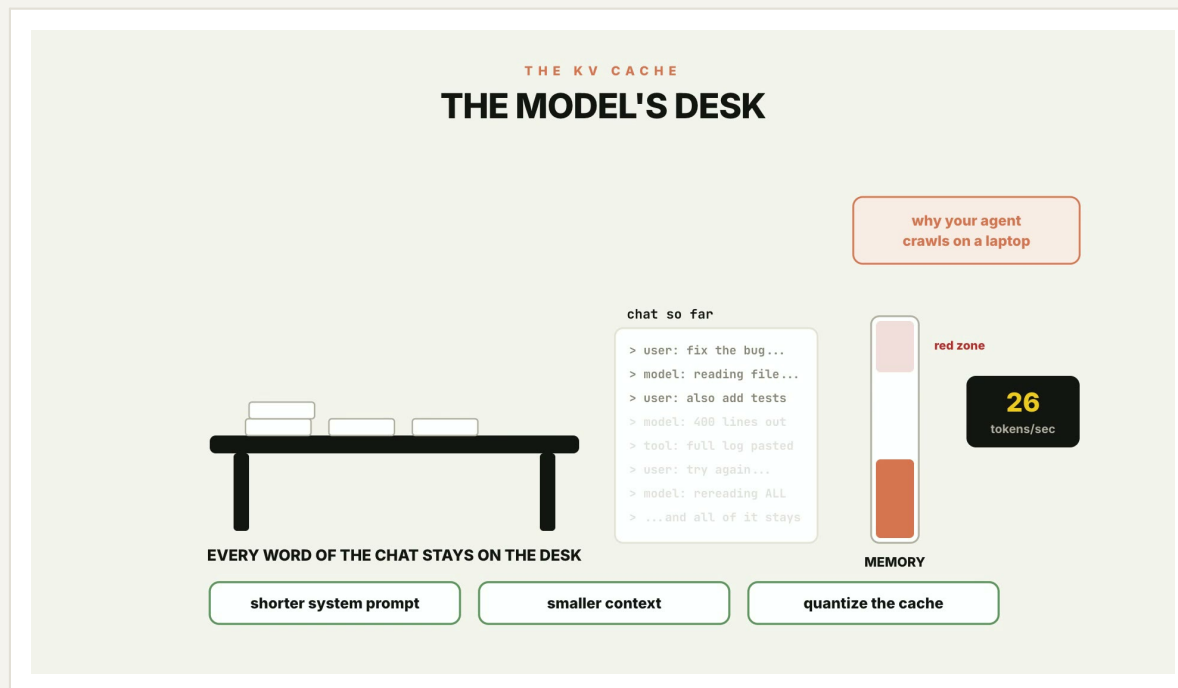
THE SIMPLE RULE

Not on Apple? GGUF, end of story. On a Mac? Look for the MLX variant first, fall back to GGUF.

AVAILABILITY

Most popular models ship both. Some only ship GGUF, which still runs fine on macOS.

The Desk Your Conversation Piles Onto



WHY CHATS SLOW DOWN

Every turn stacks context (system prompt, tool calls, history) onto the desk. The fuller the desk, the slower each reply.

PRUNE THE SYSTEM PROMPT

A bloated agent prompt turned a 2-minute 'hello' into 30 seconds once pruned. Distill what the agent actually needs.

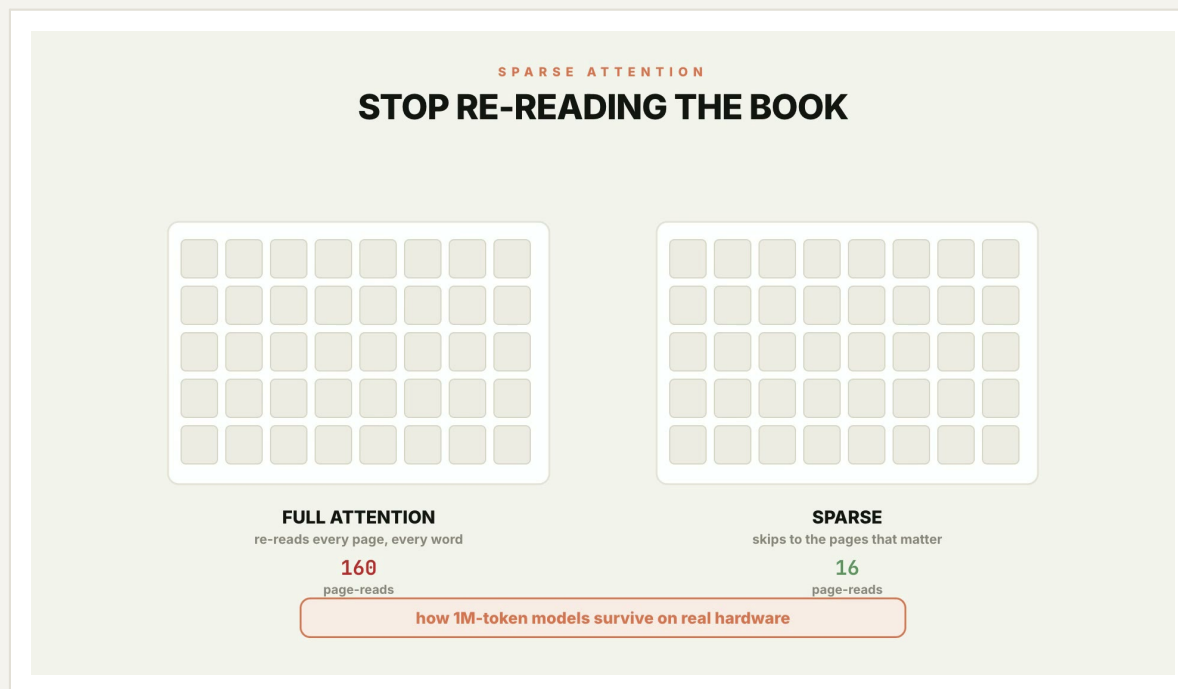
SUMMARIZE THE PAST

After ~20 messages, compress them into one and carry only the summary forward. Chat products do this constantly.

QUANTIZE THE CACHE TOO

The same compression trick that shrinks models can shrink the desk, buying your conversation more bandwidth.

Skimming Beats Re-Reading



THE STANDARD WAY

Classic attention re-reads every word of the whole conversation at every step. Thorough, and expensive.

THE SPARSE WAY

Pick and choose only the likely-relevant tokens when looking back. Dramatically more token-efficient.

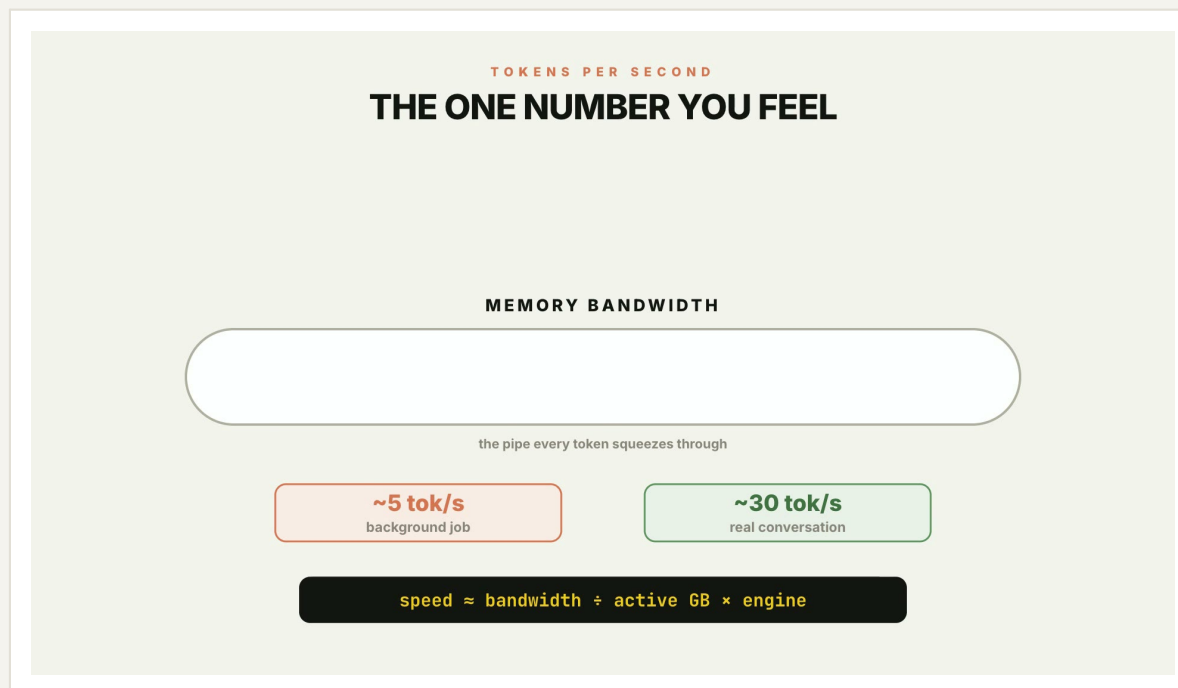
THE TRADE

Skimming can miss a line. That's the honest cost of the speedup, and for most workloads it's worth it.

WHERE YOU'LL MEET IT

Popularized by MiniMax's local models; expect the pattern to spread across open releases.

Five Is a Background Job. Thirty Is a Conversation.



THE PIPE

Memory bandwidth is a pipe; blocks of model data flow through it. Smaller (more quantized) blocks flow faster.

WHAT SLOW FEELS LIKE

5-10 tok/s is a paragraph per minute: fine for overnight agents, painful for chat.

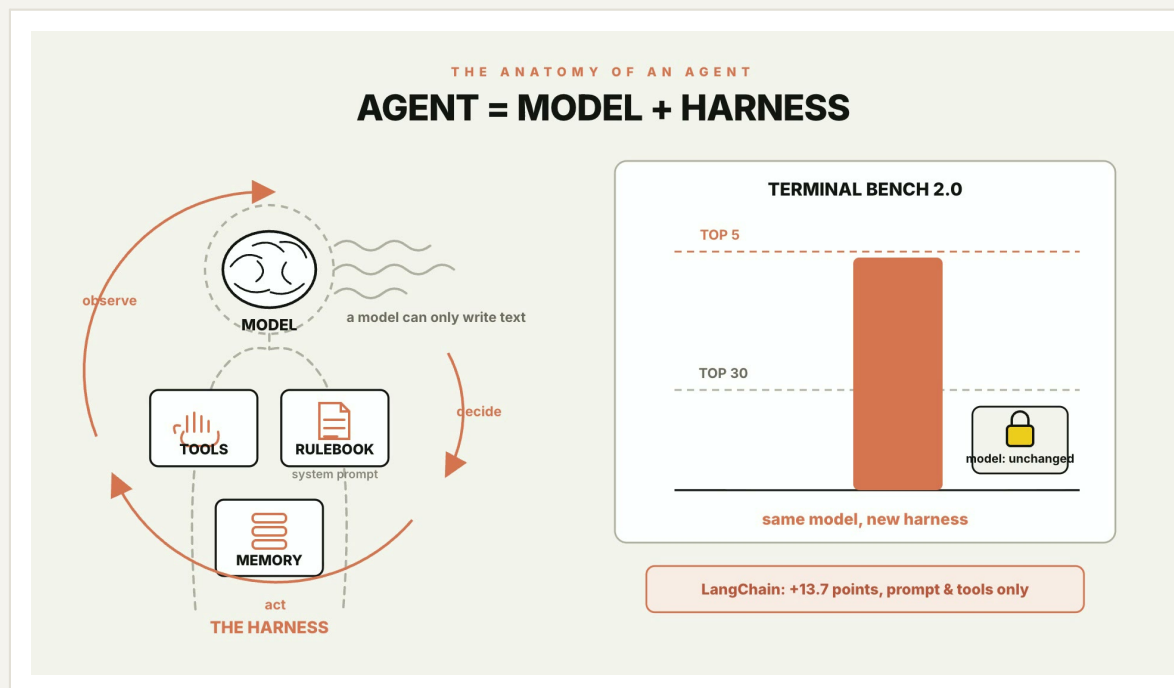
WHAT GOOD FEELS LIKE

60-70 tok/s gives you a paragraph in 10-15 seconds: a genuine conversation.

QUALITY OF LIFE

This is the number you FEEL. Pick quantizations that keep you in conversation territory on your own hardware.

A Brain in a Jar Needs Limbs



THE HARNESS ILLUSION

People credit the model for what Claude Code does. It's the symbiosis: the model plus the limbs that read, write, and run things.

BORROW A HARNESS

Hook OpenRouter (or a local server) into Claude Code or Codex and open models inherit the tools, sub-agents, and workflow you already love.

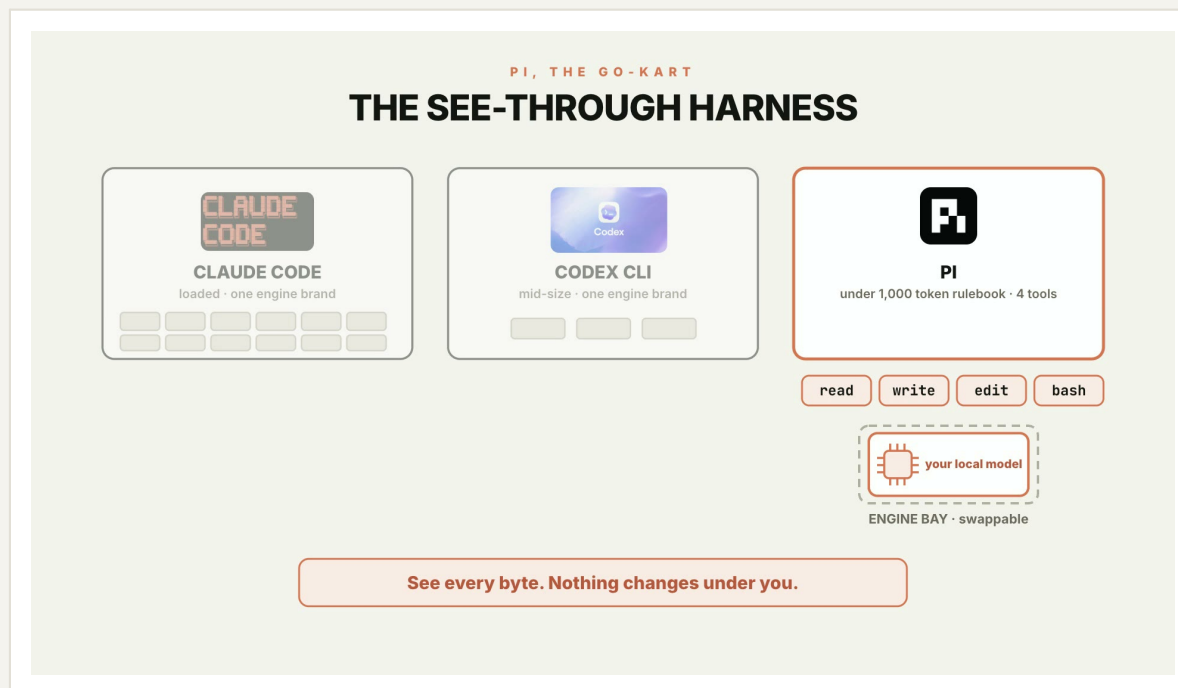
TOP 30 TO TOP 5

A mid-tier model inside a great harness leapfrogs better models running bare. Configuration is the game.

SWAP MODELS, KEEP LIMBS

One alias flips the same harness between Kimi, GLM, or whatever ships next month.

pi.dev, the Go-Kart You Own



MINIMAL AND MALLEABLE

A rulebook under a thousand tokens and four tools. Nothing hides, everything is yours to change.

IT STARTS ROUGH

Out of the box it won't match Claude Code's polish. That's expected: it hasn't been tuned to your models yet.

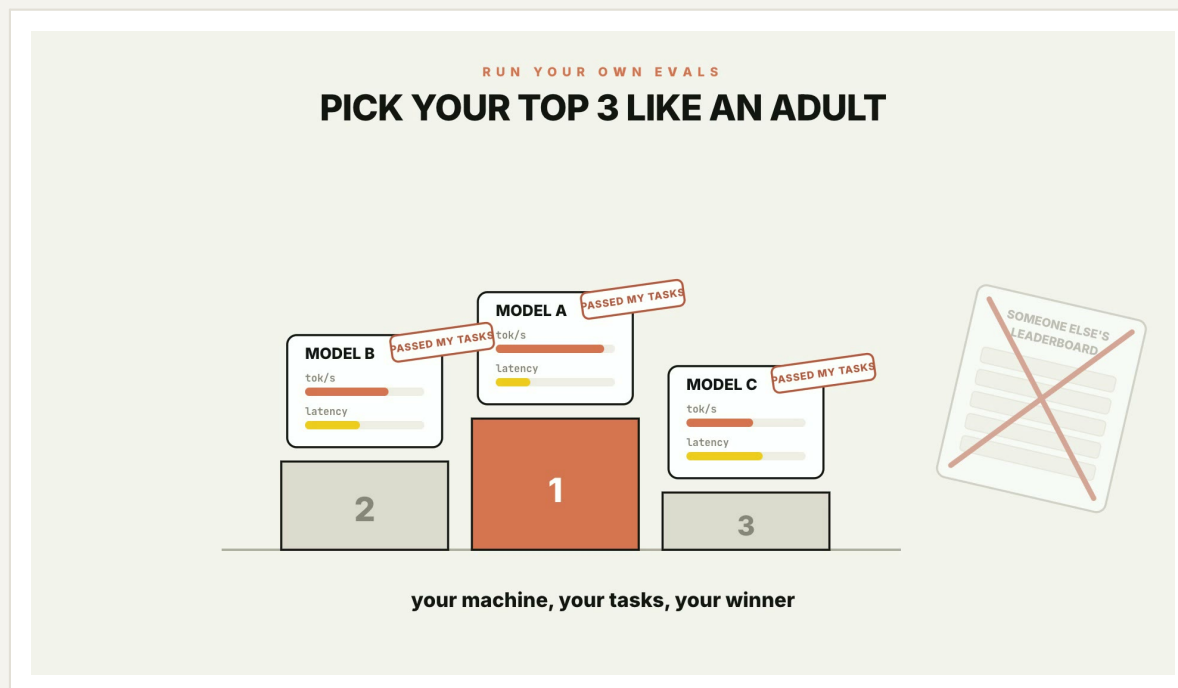
ITERATE TO OPTIMAL

Monitor tool calls, feed failures back, tweak. After 5-15 iterations it starts fitting your stack like a glove.

THE TRADE-OFF

Closed harnesses do the engineering for you, forever. An open harness needs upkeep as new models bring new habits.

The Podium Is Yours to Award



LEADERBOARDS DON'T KNOW YOU

Public benchmarks measure someone else's tasks. Distrust benchmaxxed releases; trust your own work.

THE 10-30 TASK EVAL

Collect 10-30 real tasks with answers you'd accept. Run every candidate. Score pass/fail. Done: that's an eval.

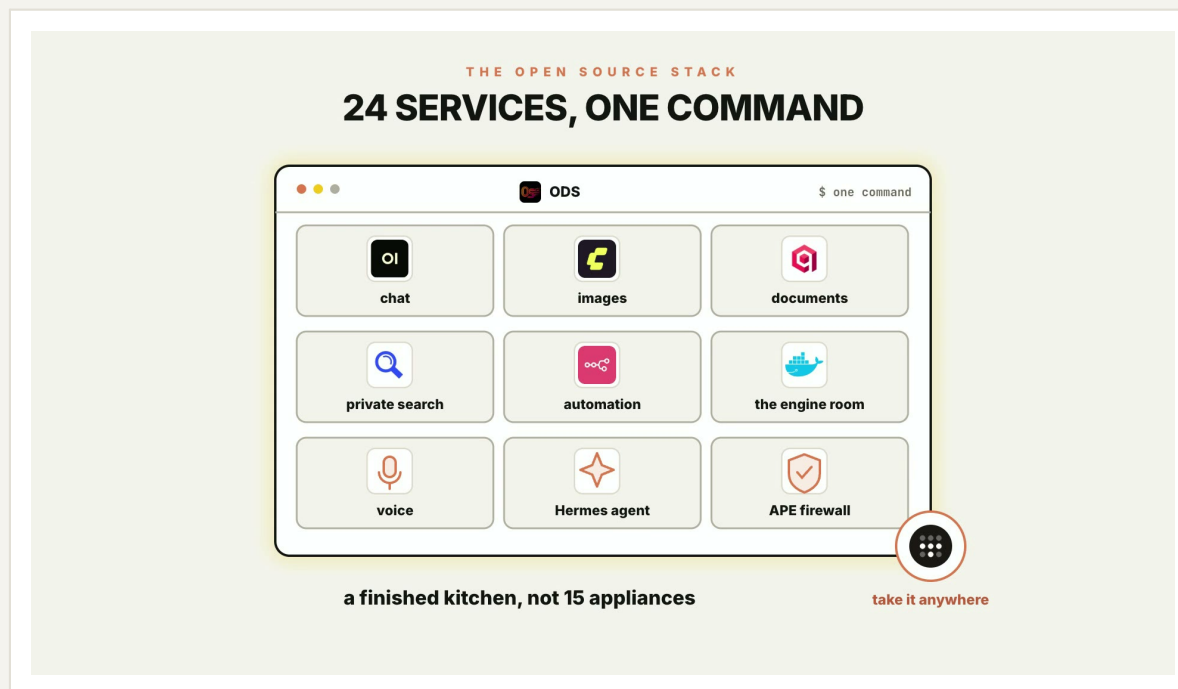
CHEAPEST PASSING MODEL WINS

The goal isn't the smartest model, it's the smallest, fastest one that clears your bar.

RE-RUN ON EVERY RELEASE

New model drops? Twenty minutes against your eval set answers whether it earns a spot on your disk.

A Finished Kitchen From One Command



THE ODS PATTERN

Take an open source command-center repo, feed it to your coding agent, have it audit, plan, and install tailored to your hardware.

WHAT YOU GET

Chat (Open WebUI), document Q&A, image generation, local n8n automations, extensions, and a live system dashboard in one place.

THE 45-MINUTE BUILD

Setup ran unattended in the background with a couple of preference questions. Plain English is the only skill required.

FRANKENSTEIN FREELY

Combine frameworks, swap models, bolt on the mesh. The perfect snowflake version of your rig is the point.

WANT TO GO DEEPER?

Level Up the Rest of Your AI Game

THE FULL LIVING COURSE

Claude Code, Codex, agentic harnesses, and how to monetize AI responsibly, taught from zero and updated as the space moves.

WALKED LINE BY LINE

The exact patterns behind builds like this local command center, broken down in dedicated modules.

DIRECT ACCESS

Me and a team of 12+ coaches when you hit a wall.

A WORKING GROUP

Builders shipping their own variants of everything you saw in this guide.

Join here: <https://www.skool.com/earlyaidopters/about>